Supplementary Material

Shortcuts and Temporal Leakage in the Bangalore EEG Epilepsy Dataset: a Provenance Audit and Leakage-Aware Benchmark of Lightweight Seizure-Type Classifiers

This document contains extended tables, additional figures, hyper-parameter grids and full per-fold results. Supplementary File S1 is the accompanying code-and-results archive (run_all.py, src/, results/, figures/, analysis.ipynb). All values were generated by that code.

S1 Hyper-parameter grids and selected values

Table S1. Hyper-parameter search grids (inner three-fold CV, macro-F1)

Model	Grid	Fixed settings
Majority	–	–
LR	C ∈ [0.01, 0.1, 1, 10, 100]	–
kNN	n_neighbors ∈ [1, 5, 15, 31]; weights ∈ ['uniform', 'distance']	–
SVM-RBF	C ∈ [1, 10, 100]; gamma ∈ ['scale', 0.01]	–
RF	max_features ∈ ['sqrt', 0.3]; min_samples_leaf ∈ [1, 5]	n_estimators=300
XGBoost	max_depth ∈ [3, 6]; learning_rate ∈ [0.05, 0.2]	colsample_bytree=0.8; n_estimators=300; subsample=0.8
LightGBM	num_leaves ∈ [15, 63]; learning_rate ∈ [0.05, 0.2]	colsample_bytree=0.8; n_estimators=300; subsample=0.8
MLP	– (fixed architecture)	AdamW lr 0.002, weight decay 1e-4, batch 128, ≤60 epochs, patience 8
1D-CNN	– (fixed architecture)	AdamW lr 0.002, weight decay 1e-4, batch 128, ≤60 epochs, patience 8
BiLSTM	– (fixed architecture)	AdamW lr 0.002, weight decay 1e-4, batch 128, ≤60 epochs, patience 8
Transformer	– (fixed architecture)	AdamW lr 0.002, weight decay 1e-4, batch 128, ≤60 epochs, patience 8

Table S2. Hyper-parameters selected on the development set for each hold-out run

Task	Protocol	Features	Model	Selected
4class	random	raw	LR	{logisticregression__C: 0.01}
4class	random	raw	kNN	{kneighborsclassifier__n_neighbors: 1, kneighborsclassifier__weights: uniform}
4class	random	raw	SVM-RBF	{svc__C: 100, svc__gamma: scale}
4class	random	raw	RF	{max_features: sqrt, min_samples_leaf: 1}
4class	random	raw	XGBoost	{learning_rate: 0.2, max_depth: 6}
4class	random	raw	LightGBM	{learning_rate: 0.2, num_leaves: 63}
4class	random	eng	LR	{logisticregression__C: 10}
4class	random	eng	kNN	{kneighborsclassifier__n_neighbors: 1, kneighborsclassifier__weights: uniform}
4class	random	eng	SVM-RBF	{svc__C: 100, svc__gamma: scale}
4class	random	eng	RF	{max_features: 0.3, min_samples_leaf: 1}
4class	random	eng	XGBoost	{learning_rate: 0.2, max_depth: 6}
4class	random	eng	LightGBM	{learning_rate: 0.05, num_leaves: 63}

4class	random	raw+eng	LR	{logisticregression__C: 100}
4class	random	raw+eng	kNN	{kneighborsclassifier__n_neighbors: 1, kneighborsclassifier__weights: uniform}
4class	random	raw+eng	SVM-RBF	{svc__C: 100, svc__gamma: scale}
4class	random	raw+eng	RF	{max_features: 0.3, min_samples_leaf: 1}
4class	random	raw+eng	XGBoost	{learning_rate: 0.2, max_depth: 6}
4class	random	raw+eng	LightGBM	{learning_rate: 0.05, num_leaves: 63}
4class	blocked	raw	LR	{logisticregression__C: 100}
4class	blocked	raw	kNN	{kneighborsclassifier__n_neighbors: 5, kneighborsclassifier__weights: uniform}
4class	blocked	raw	SVM-RBF	{svc__C: 100, svc__gamma: scale}
4class	blocked	raw	RF	{max_features: sqrt, min_samples_leaf: 1}
4class	blocked	raw	XGBoost	{learning_rate: 0.2, max_depth: 3}
4class	blocked	raw	LightGBM	{learning_rate: 0.05, num_leaves: 15}
4class	blocked	eng	LR	{logisticregression__C: 100}
4class	blocked	eng	kNN	{kneighborsclassifier__n_neighbors: 15, kneighborsclassifier__weights: distance}
4class	blocked	eng	SVM-RBF	{svc__C: 100, svc__gamma: 0.01}
4class	blocked	eng	RF	{max_features: sqrt, min_samples_leaf: 1}
4class	blocked	eng	XGBoost	{learning_rate: 0.2, max_depth: 3}
4class	blocked	eng	LightGBM	{learning_rate: 0.2, num_leaves: 15}
4class	blocked	raw+eng	LR	{logisticregression__C: 100}
4class	blocked	raw+eng	kNN	{kneighborsclassifier__n_neighbors: 15, kneighborsclassifier__weights: distance}
4class	blocked	raw+eng	SVM-RBF	{svc__C: 10, svc__gamma: scale}
4class	blocked	raw+eng	RF	{max_features: 0.3, min_samples_leaf: 1}
4class	blocked	raw+eng	XGBoost	{learning_rate: 0.2, max_depth: 3}
4class	blocked	raw+eng	LightGBM	{learning_rate: 0.2, num_leaves: 15}
binary	random	raw+eng	LR	{logisticregression__C: 1}
binary	random	raw+eng	kNN	{kneighborsclassifier__n_neighbors: 1, kneighborsclassifier__weights: uniform}
binary	random	raw+eng	SVM-RBF	{svc__C: 1, svc__gamma: scale}
binary	random	raw+eng	RF	{max_features: sqrt, min_samples_leaf: 1}
binary	random	raw+eng	XGBoost	{learning_rate: 0.05, max_depth: 3}
binary	random	raw+eng	LightGBM	{learning_rate: 0.05, num_leaves: 15}
binary	blocked	raw+eng	LR	{logisticregression__C: 10}
binary	blocked	raw+eng	kNN	{kneighborsclassifier__n_neighbors: 5, kneighborsclassifier__weights: uniform}
binary	blocked	raw+eng	SVM-RBF	{svc__C: 10, svc__gamma: 0.01}
binary	blocked	raw+eng	RF	{max_features: sqrt, min_samples_leaf: 1}
binary	blocked	raw+eng	XGBoost	{learning_rate: 0.05,

				max_depth: 3}
binary	blocked	raw+eng	LightGBM	{learning_rate: 0.05, num_leaves: 15}
3class	random	raw+eng	LR	{logisticregression__C: 100}
3class	random	raw+eng	kNN	{kneighborsclassifier__n_neighbors: 1, kneighborsclassifier__weights: uniform}
3class	random	raw+eng	SVM-RBF	{svc__C: 100, svc__gamma: scale}
3class	random	raw+eng	RF	{max_features: 0.3, min_samples_leaf: 1}
3class	random	raw+eng	XGBoost	{learning_rate: 0.2, max_depth: 6}
3class	random	raw+eng	LightGBM	{learning_rate: 0.05, num_leaves: 63}
3class	blocked	raw+eng	LR	{logisticregression__C: 1}
3class	blocked	raw+eng	kNN	{kneighborsclassifier__n_neighbors: 15, kneighborsclassifier__weights: uniform}
3class	blocked	raw+eng	SVM-RBF	{svc__C: 10, svc__gamma: 0.01}
3class	blocked	raw+eng	RF	{max_features: 0.3, min_samples_leaf: 1}
3class	blocked	raw+eng	XGBoost	{learning_rate: 0.2, max_depth: 3}
3class	blocked	raw+eng	LightGBM	{learning_rate: 0.2, num_leaves: 15}

S2 Extended hold-out results

Table S3. Four-class hold-out results for all models and feature sets

Protocol	Features	Model	Accuracy	Macro-F1	Kappa	AUC	Brier	ECE	Fit (s)
random	eng	LightGBM	0.916	0.916	0.888	0.988	0.128	0.025	10.9
random	eng	XGBoost	0.909	0.909	0.878	0.985	0.140	0.019	8.8
random	eng	kNN	0.902	0.903	0.870	0.935	0.195	0.098	0.4
random	eng	SVM-RBF	0.889	0.890	0.853	0.979	0.165	0.034	14.5
random	eng	RF	0.871	0.871	0.828	0.978	0.212	0.114	24.3
random	eng	LR	0.749	0.749	0.666	0.925	0.323	0.033	4.3
random	eng	Majority	0.250	0.100	0.000	0.500	0.750	0.000	0.0
random	raw	kNN	0.986	0.986	0.982	0.991	0.028	0.014	0.2
random	raw	LightGBM	0.986	0.986	0.981	0.999	0.025	0.010	3.4
random	raw	XGBoost	0.976	0.976	0.968	0.999	0.035	0.009	2.5
random	raw	RF	0.964	0.964	0.953	0.998	0.096	0.133	7.5
random	raw	1D-CNN	0.956	0.956	0.941	0.997	0.066	0.015	9.0
random	raw	MLP	0.932	0.933	0.910	0.994	0.091	0.029	7.1
random	raw	Transformer	0.921	0.921	0.895	0.988	0.121	0.018	32.5
random	raw	SVM-RBF	0.903	0.903	0.871	0.983	0.145	0.027	6.7
random	raw	BiLSTM	0.706	0.697	0.607	0.893	0.374	0.029	9.0
random	raw	LR	0.477	0.483	0.302	0.657	0.680	0.127	2.8
random	raw	Majority	0.250	0.100	0.000	0.500	0.750	0.000	0.0
random	raw+eng	LightGBM	0.973	0.973	0.964	0.999	0.039	0.010	14.8
random	raw+eng	XGBoost	0.962	0.962	0.949	0.997	0.053	0.009	9.1
random	raw+eng	RF	0.943	0.943	0.924	0.994	0.135	0.136	28.3
random	raw+eng	SVM-RBF	0.928	0.928	0.904	0.990	0.111	0.036	13.6
random	raw+eng	kNN	0.926	0.926	0.902	0.951	0.147	0.074	0.4
random	raw+eng	LR	0.761	0.760	0.681	0.931	0.311	0.030	15.8
random	raw+eng	Majority	0.250	0.100	0.000	0.500	0.750	0.000	0.0
blocked	eng	LR	0.707	0.705	0.609	0.909	0.357	0.054	5.8
blocked	eng	SVM-RBF	0.702	0.697	0.603	0.891	0.426	0.124	10.7
blocked	eng	LightGBM	0.699	0.690	0.599	0.889	0.497	0.208	7.0
blocked	eng	XGBoost	0.695	0.686	0.593	0.894	0.419	0.129	5.9
blocked	eng	RF	0.686	0.664	0.581	0.882	0.393	0.029	15.6
blocked	eng	kNN	0.649	0.631	0.532	0.847	0.462	0.094	0.3
blocked	eng	Majority	0.250	0.100	0.000	0.500	0.750	0.007	0.0
blocked	raw	SVM-RBF	0.844	0.842	0.792	0.962	0.238	0.053	5.4
blocked	raw	LightGBM	0.816	0.810	0.755	0.949	0.276	0.077	2.9
blocked	raw	XGBoost	0.805	0.798	0.740	0.946	0.293	0.094	1.8
blocked	raw	RF	0.803	0.795	0.737	0.942	0.287	0.077	6.6
blocked	raw	MLP	0.779	0.772	0.705	0.949	0.309	0.076	2.4
blocked	raw	Transformer	0.752	0.746	0.669	0.938	0.335	0.096	11.6

blocked	raw	1D-CNN	0.725	0.722	0.633	0.926	0.373	0.132	3.6
blocked	raw	kNN	0.709	0.710	0.611	0.861	0.443	0.160	0.2
blocked	raw	BiLSTM	0.520	0.468	0.360	0.771	0.546	0.093	2.2
blocked	raw	LR	0.440	0.436	0.253	0.666	0.674	0.087	0.5
blocked	raw	Majority	0.250	0.100	0.000	0.500	0.750	0.007	0.0
blocked	raw+eng	LightGBM	0.807	0.799	0.742	0.942	0.341	0.148	8.4
blocked	raw+eng	XGBoost	0.790	0.784	0.720	0.940	0.310	0.096	6.6
blocked	raw+eng	RF	0.768	0.755	0.691	0.919	0.329	0.033	25.9
blocked	raw+eng	LR	0.724	0.724	0.631	0.915	0.350	0.058	12.1
blocked	raw+eng	SVM-RBF	0.728	0.719	0.637	0.912	0.378	0.103	8.7
blocked	raw+eng	kNN	0.656	0.633	0.541	0.863	0.435	0.082	0.4
blocked	raw+eng	Majority	0.250	0.100	0.000	0.500	0.750	0.007	0.0

Fit time includes nested hyper-parameter search on 8 CPU cores.

Table S4. Row-level versus cluster bootstrap 95% confidence intervals (four-class)

Protocol	Features	Model	Macro-F1	Row CI	Cluster CI	Accuracy	Row CI	Cluster CI
random	raw	kNN	0.986	0.981–0.992	0.980–0.992	0.986	0.981–0.992	0.980–0.992
random	raw	LightGBM	0.986	0.979–0.991	0.979–0.992	0.986	0.979–0.991	0.979–0.992
random	raw	XGBoost	0.976	0.969–0.984	0.968–0.983	0.976	0.969–0.984	0.968–0.984
random	raw+eng	LightGBM	0.973	0.966–0.981	0.964–0.982	0.973	0.966–0.981	0.964–0.982
random	raw	RF	0.964	0.956–0.974	0.953–0.974	0.964	0.956–0.974	0.953–0.974
random	raw+eng	XGBoost	0.962	0.953–0.971	0.951–0.972	0.962	0.953–0.971	0.950–0.972
random	raw	1D-CNN	0.956	0.945–0.966	0.945–0.966	0.956	0.945–0.966	0.945–0.966
random	raw+eng	RF	0.943	0.931–0.954	0.929–0.956	0.943	0.931–0.954	0.929–0.956
random	raw	MLP	0.933	0.921–0.945	0.916–0.949	0.932	0.920–0.945	0.915–0.949
random	raw+eng	SVM-RBF	0.928	0.916–0.941	0.913–0.944	0.928	0.915–0.941	0.912–0.944
random	raw+eng	kNN	0.926	0.914–0.938	0.912–0.939	0.926	0.914–0.938	0.912–0.939
random	raw	Transformer	0.921	0.908–0.933	0.901–0.939	0.921	0.909–0.933	0.902–0.939
random	raw	SVM-RBF	0.903	0.888–0.918	0.880–0.924	0.903	0.888–0.917	0.879–0.924
random	raw+eng	LR	0.760	0.740–0.780	0.736–0.786	0.761	0.741–0.781	0.735–0.787
random	raw	BiLSTM	0.697	0.677–0.717	0.664–0.730	0.706	0.687–0.725	0.671–0.741
random	raw	LR	0.483	0.460–0.506	0.444–0.518	0.477	0.454–0.501	0.438–0.513
blocked	raw	SVM-RBF	0.842	0.824–0.859	0.800–0.880	0.844	0.827–0.861	0.802–0.882
blocked	raw	LightGBM	0.810	0.791–0.827	0.771–0.845	0.816	0.799–0.832	0.779–0.850
blocked	raw+eng	LightGBM	0.799	0.781–0.817	0.754–0.843	0.807	0.790–0.824	0.766–0.847
blocked	raw	XGBoost	0.798	0.780–0.816	0.755–0.838	0.805	0.788–0.822	0.764–0.842
blocked	raw	RF	0.795	0.777–0.813	0.736–0.848	0.803	0.786–0.821	0.750–0.853
blocked	raw+eng	XGBoost	0.784	0.765–0.801	0.740–0.826	0.790	0.772–0.807	0.748–0.830
blocked	raw	MLP	0.772	0.752–0.790	0.720–0.820	0.779	0.761–0.796	0.730–0.825
blocked	raw+eng	RF	0.755	0.737–0.774	0.702–0.806	0.768	0.751–0.786	0.725–0.812
blocked	raw	Transformer	0.746	0.726–0.765	0.699–0.790	0.752	0.732–0.770	0.708–0.794
blocked	raw+eng	LR	0.724	0.705–0.744	0.672–0.775	0.724	0.704–0.744	0.673–0.775
blocked	raw	1D-CNN	0.722	0.701–0.742	0.674–0.765	0.725	0.705–0.744	0.680–0.767
blocked	raw+eng	SVM-RBF	0.719	0.699–0.738	0.674–0.769	0.728	0.710–0.747	0.686–0.774
blocked	raw	kNN	0.710	0.690–0.731	0.667–0.750	0.709	0.688–0.729	0.667–0.746
blocked	raw+eng	kNN	0.633	0.613–0.651	0.601–0.668	0.656	0.638–0.674	0.622–0.692
blocked	raw	BiLSTM	0.468	0.453–0.482	0.410–0.526	0.520	0.503–0.536	0.452–0.589
blocked	raw	LR	0.436	0.414–0.458	0.369–0.500	0.440	0.417–0.462	0.370–0.506

Row CI: independent resampling of test rows. Cluster CI: class-stratified resampling of Bonn segments (class 0) and 50-row chunks (classes 1–3), which respects autocorrelation and is the primary interval in the paper.

Table S5. Per-class precision, recall and F1 of the best hold-out models

Task	Protocol	Model	Class	Precision	Recall	F1	Support
4class	random	kNN (raw)	H	1.000	0.988	0.994	400
4class	random	kNN (raw)	GS	0.992	0.983	0.987	400
4class	random	kNN (raw)	FS	0.987	0.975	0.981	400
4class	random	kNN (raw)	SE	0.966	1.000	0.983	400
4class	blocked	SVM-RBF (raw)	H	1.000	0.995	0.997	399
4class	blocked	SVM-RBF (raw)	GS	0.799	0.912	0.852	400
4class	blocked	SVM-RBF (raw)	FS	0.809	0.802	0.806	400
4class	blocked	SVM-RBF (raw)	SE	0.764	0.665	0.711	400
3class	random	kNN (raw)	GS	0.995	0.985	0.990	400
3class	random	kNN (raw)	FS	0.990	0.985	0.987	400
3class	random	kNN (raw)	SE	0.985	1.000	0.993	400
3class	blocked	SVM-RBF (raw)	GS	0.760	0.925	0.834	400
3class	blocked	SVM-RBF (raw)	FS	0.824	0.807	0.816	400
3class	blocked	SVM-RBF (raw)	SE	0.760	0.610	0.677	400

Table S6. Binary and three-class hold-out results

Task	Protocol	Features	Model	Accuracy	Macro-F1	Cluster CI	Kappa	AUC	ECE
binary	random	raw	1D-CNN	1.000	1.000	1.000–1.000	1.000	1.000	0.003
binary	random	raw+eng	XGBoost	0.999	0.999	0.997–1.000	0.998	1.000	0.001
binary	random	raw+eng	LR	0.999	0.999	0.997–1.000	0.998	1.000	0.001
binary	random	raw+eng	RF	0.999	0.999	0.997–1.000	0.998	1.000	0.001
binary	random	raw+eng	SVM-RBF	0.999	0.998	0.996–1.000	0.997	1.000	0.001
binary	random	raw+eng	LightGBM	0.999	0.998	0.996–1.000	0.997	1.000	0.001
binary	random	raw+eng	kNN	0.999	0.998	0.996–1.000	0.997	0.998	0.001
binary	random	raw	MLP	0.998	0.997	0.994–1.000	0.995	1.000	0.001
binary	random	raw+eng	Majority	0.750	0.429	0.422–0.434	0.000	0.500	0.000
binary	blocked	raw+eng	LR	0.999	0.999	0.997–1.000	0.998	1.000	0.001
binary	blocked	raw	1D-CNN	0.999	0.999	0.997–1.000	0.998	1.000	0.001
binary	blocked	raw+eng	SVM-RBF	0.999	0.999	0.997–1.000	0.998	1.000	0.001
binary	blocked	raw+eng	RF	0.999	0.999	0.997–1.000	0.998	1.000	0.002
binary	blocked	raw+eng	LightGBM	0.999	0.999	0.997–1.000	0.998	1.000	0.001
binary	blocked	raw+eng	XGBoost	0.999	0.998	0.996–1.000	0.997	1.000	0.001
binary	blocked	raw+eng	kNN	0.999	0.998	0.995–1.000	0.997	0.999	0.001
binary	blocked	raw	MLP	0.996	0.995	0.989–0.999	0.990	1.000	0.010
binary	blocked	raw+eng	Majority	0.750	0.429	0.426–0.431	0.000	0.500	0.007
3class	random	raw	kNN	0.990	0.990	0.980–0.997	0.985	0.993	0.010
3class	random	raw	LightGBM	0.968	0.968	0.954–0.980	0.951	0.997	0.012
3class	random	raw	SVM-RBF	0.964	0.964	0.950–0.977	0.946	0.996	0.025
3class	random	raw+eng	LightGBM	0.960	0.960	0.944–0.974	0.940	0.996	0.014
3class	random	raw+eng	kNN	0.959	0.959	0.946–0.971	0.939	0.969	0.041
3class	random	raw	XGBoost	0.958	0.958	0.945–0.971	0.938	0.995	0.009
3class	random	raw+eng	XGBoost	0.956	0.956	0.939–0.971	0.934	0.994	0.010
3class	random	raw+eng	SVM-RBF	0.955	0.955	0.941–0.968	0.932	0.993	0.015
3class	random	raw	RF	0.948	0.948	0.930–0.963	0.921	0.995	0.151
3class	random	raw	MLP	0.944	0.944	0.926–0.962	0.916	0.992	0.022
3class	random	raw	1D-CNN	0.931	0.931	0.914–0.948	0.896	0.989	0.027
3class	random	raw+eng	RF	0.924	0.924	0.903–0.944	0.886	0.987	0.174
3class	random	raw+eng	LR	0.721	0.719	0.681–0.753	0.581	0.875	0.032
3class	random	raw	LR	0.491	0.489	0.446–0.533	0.236	0.663	0.045
3class	random	raw+eng	Majority	0.333	0.167	0.158–0.175	0.000	0.500	0.000
3class	random	raw	Majority	0.333	0.167	0.158–0.175	0.000	0.500	0.000
3class	blocked	raw	SVM-RBF	0.781	0.776	0.712–0.832	0.671	0.923	0.093
3class	blocked	raw	MLP	0.776	0.770	0.705–0.830	0.664	0.925	0.075
3class	blocked	raw	LightGBM	0.762	0.752	0.694–0.806	0.644	0.903	0.179
3class	blocked	raw	XGBoost	0.742	0.728	0.670–0.779	0.613	0.898	0.163

3class	blocked	raw+eng	LightGBM	0.734	0.722	0.662–0.776	0.601	0.888	0.200
3class	blocked	raw	RF	0.730	0.718	0.644–0.779	0.595	0.889	0.089
3class	blocked	raw+eng	XGBoost	0.719	0.711	0.650–0.766	0.579	0.874	0.137
3class	blocked	raw	1D-CNN	0.714	0.706	0.657–0.749	0.571	0.878	0.127
3class	blocked	raw+eng	RF	0.679	0.661	0.581–0.732	0.519	0.840	0.035
3class	blocked	raw+eng	SVM-RBF	0.675	0.657	0.610–0.701	0.512	0.850	0.183
3class	blocked	raw	kNN	0.647	0.646	0.577–0.706	0.470	0.820	0.091
3class	blocked	raw+eng	LR	0.637	0.637	0.564–0.704	0.456	0.832	0.063
3class	blocked	raw+eng	kNN	0.603	0.574	0.530–0.617	0.404	0.776	0.089
3class	blocked	raw	LR	0.441	0.438	0.355–0.514	0.161	0.630	0.066
3class	blocked	raw+eng	Majority	0.333	0.167	0.167–0.167	0.000	0.500	0.000
3class	blocked	raw	Majority	0.333	0.167	0.167–0.167	0.000	0.500	0.000

Table S7. Single-feature, single-threshold classifiers for healthy versus seizure

Protocol	Feature	Threshold	Accuracy	Macro-F1
random	t_lag1_ac	0.372	0.997	0.996
random	t_std	21.434	0.943	0.922
random	f_low_high_ratio	1.749	0.996	0.995
random	t_zc_mean	4.500	0.985	0.980
blocked	t_lag1_ac	0.369	0.997	0.997
blocked	t_std	21.568	0.955	0.939
blocked	f_low_high_ratio	1.450	0.993	0.991
blocked	t_zc_mean	4.500	0.984	0.978

Depth-1 decision tree trained on the development set; evaluated on the hold-out test set.

S3 Cross-validation, statistics and leakage diagnostics

Table S8. Per-fold macro-F1, nested 5 × 2 CV, 4class, random protocol

Fold	1D-CNN	LR	LightGBM	MLP	RF	SVM-RBF	XGBoost	kNN
1	0.938	0.768	0.963	0.916	0.934	0.901	0.955	0.894
2	0.951	0.761	0.967	0.946	0.946	0.917	0.959	0.899
3	0.921	0.776	0.967	0.906	0.927	0.915	0.956	0.898
4	0.902	0.768	0.967	0.926	0.932	0.910	0.968	0.894
5	0.930	0.791	0.969	0.924	0.944	0.920	0.965	0.912
6	0.946	0.772	0.970	0.910	0.942	0.916	0.961	0.891
7	0.930	0.756	0.956	0.921	0.910	0.897	0.949	0.906
8	0.939	0.796	0.968	0.938	0.945	0.928	0.961	0.914
9	0.934	0.782	0.964	0.909	0.945	0.919	0.955	0.907
10	0.946	0.775	0.962	0.934	0.928	0.906	0.961	0.914
Mean	0.934	0.775	0.965	0.923	0.935	0.913	0.959	0.903

Table S9. Per-fold macro-F1, nested 5 × 2 CV, 4class, blocked protocol

Fold	1D-CNN	LR	LightGBM	MLP	RF	SVM-RBF	XGBoost	kNN
1	0.669	0.611	0.657	0.646	0.644	0.617	0.660	0.583
2	0.735	0.611	0.750	0.632	0.686	0.625	0.723	0.624
3	0.711	0.665	0.791	0.771	0.760	0.668	0.785	0.620
4	0.684	0.611	0.765	0.683	0.698	0.608	0.747	0.582
5	0.775	0.680	0.819	0.804	0.772	0.679	0.813	0.649
6	0.739	0.646	0.768	0.780	0.736	0.648	0.760	0.635
7	0.752	0.679	0.803	0.804	0.731	0.745	0.804	0.680
8	0.717	0.656	0.699	0.739	0.655	0.642	0.717	0.604
9	0.824	0.722	0.852	0.798	0.799	0.753	0.833	0.688
10	0.723	0.654	0.762	0.785	0.757	0.660	0.766	0.627
Mean	0.733	0.653	0.767	0.744	0.724	0.665	0.761	0.629

Table S10. Per-fold macro-F1, nested 5 × 2 CV, 3class, random protocol

Fold	1D-CNN	LR	LightGBM	MLP	RF	SVM-RBF	XGBoost	kNN
1	0.927	0.684	0.956	0.918	0.905	0.932	0.944	0.926
2	0.941	0.701	0.961	0.942	0.915	0.947	0.955	0.932
3	0.933	0.712	0.966	0.944	0.921	0.945	0.951	0.933
4	0.915	0.675	0.950	0.933	0.909	0.949	0.943	0.944
5	0.937	0.702	0.969	0.957	0.934	0.950	0.959	0.948
6	0.929	0.703	0.952	0.927	0.907	0.944	0.938	0.942
7	0.915	0.699	0.964	0.937	0.937	0.944	0.954	0.937
8	0.937	0.693	0.952	0.936	0.924	0.942	0.951	0.936
9	0.948	0.691	0.965	0.946	0.917	0.938	0.954	0.933
10	0.915	0.683	0.955	0.938	0.916	0.948	0.942	0.943
Mean	0.930	0.694	0.959	0.938	0.918	0.944	0.949	0.937

Table S11. Per-fold macro-F1, nested 5 × 2 CV, 3class, blocked protocol

Fold	1D-CNN	LR	LightGBM	MLP	RF	SVM-RBF	XGBoost	kNN
1	0.537	0.484	0.635	0.521	0.568	0.543	0.643	0.517
2	0.597	0.574	0.661	0.671	0.585	0.645	0.666	0.573
3	0.674	0.515	0.696	0.573	0.659	0.665	0.696	0.581
4	0.684	0.586	0.721	0.687	0.679	0.623	0.709	0.518
5	0.731	0.566	0.737	0.751	0.608	0.613	0.752	0.534
6	0.578	0.459	0.690	0.611	0.583	0.482	0.640	0.458
7	0.544	0.449	0.560	0.581	0.504	0.485	0.529	0.474
8	0.718	0.592	0.737	0.724	0.650	0.599	0.732	0.560
9	0.833	0.628	0.848	0.843	0.805	0.780	0.849	0.708
10	0.562	0.590	0.677	0.643	0.602	0.583	0.698	0.539
Mean	0.646	0.544	0.696	0.661	0.624	0.602	0.692	0.546

Table S12. Corrected resampled t-tests against the best model (all settings)

Task	Protocol	Best	Other	F1 best	F1 other	t	p	p (Holm)
3class	blocked	LightGBM	1D-CNN	0.696	0.646	1.89	0.0911	0.2734
3class	blocked	LightGBM	LR	0.696	0.544	4.92	0.0008	0.0058
3class	blocked	LightGBM	MLP	0.696	0.661	1.10	0.2991	0.5983
3class	blocked	LightGBM	RF	0.696	0.624	3.91	0.0035	0.0177
3class	blocked	LightGBM	SVM-RBF	0.696	0.602	2.81	0.0205	0.0821
3class	blocked	LightGBM	XGBoost	0.696	0.692	0.35	0.7328	0.7328
3class	blocked	LightGBM	kNN	0.696	0.546	4.78	0.0010	0.0060
3class	random	LightGBM	1D-CNN	0.959	0.930	4.69	0.0011	0.0046
3class	random	LightGBM	LR	0.959	0.694	50.35	0.0000	0.0000
3class	random	LightGBM	MLP	0.959	0.938	4.90	0.0008	0.0042
3class	random	LightGBM	RF	0.959	0.918	8.16	0.0000	0.0001
3class	random	LightGBM	SVM-RBF	0.959	0.944	3.12	0.0123	0.0123
3class	random	LightGBM	XGBoost	0.959	0.949	3.93	0.0035	0.0104
3class	random	LightGBM	kNN	0.959	0.937	3.82	0.0041	0.0104
4class	blocked	LightGBM	1D-CNN	0.767	0.733	1.65	0.1331	0.3994
4class	blocked	LightGBM	LR	0.767	0.653	4.88	0.0009	0.0053
4class	blocked	LightGBM	MLP	0.767	0.744	0.75	0.4720	0.9440
4class	blocked	LightGBM	RF	0.767	0.724	3.08	0.0130	0.0522
4class	blocked	LightGBM	SVM-RBF	0.767	0.665	4.34	0.0019	0.0094
4class	blocked	LightGBM	XGBoost	0.767	0.761	0.72	0.4887	0.9440
4class	blocked	LightGBM	kNN	0.767	0.629	6.44	0.0001	0.0008
4class	random	LightGBM	1D-CNN	0.965	0.934	3.62	0.0055	0.0111
4class	random	LightGBM	LR	0.965	0.775	29.96	0.0000	0.0000
4class	random	LightGBM	MLP	0.965	0.923	5.06	0.0007	0.0020
4class	random	LightGBM	RF	0.965	0.935	5.70	0.0003	0.0012
4class	random	LightGBM	SVM-RBF	0.965	0.913	12.87	0.0000	0.0000
4class	random	LightGBM	XGBoost	0.965	0.959	2.62	0.0278	0.0278
4class	random	LightGBM	kNN	0.965	0.903	9.98	0.0000	0.0000

Table S13. Friedman tests and mean ranks (Nemenyi critical difference)

Task	Protocol	χ^2	p	CD	Mean ranks
3class	blocked	55.50	1.19e-09	3.32	LightGBM 1.80; XGBoost 1.80; MLP 3.20; 1D-CNN 4.40; RF 4.80; SVM-RBF 5.30; LR 7.20; kNN 7.50
3class	random	60.40	1.26e-10	3.32	LightGBM 1.00; XGBoost 2.60; SVM-RBF 3.00; kNN 4.60; MLP 4.80; 1D-CNN 5.20; RF 6.80; LR 8.00

4class	blocked	57.23	5.37e-10	3.32	LightGBM 2.00; XGBoost 2.10; MLP 3.00; 1D-CNN 3.50; RF 4.60; SVM-RBF 6.20; LR 6.80; kNN 7.80
4class	random	64.73	1.70e-11	3.32	LightGBM 1.10; XGBoost 1.90; 1D-CNN 3.70; RF 3.80; MLP 4.90; SVM-RBF 5.80; kNN 6.80; LR 8.00

Folds of repeated CV are not independent, so Friedman/Nemenyi results are indicative only.

Table S14. Nearest training neighbour of random-split test rows

Class	Same class	Within 1 row	Within 5 rows	Median distance (rows)
0	0.988	0.000	0.007	643
1	0.983	0.922	0.975	1
2	0.975	0.877	0.932	1
3	1.000	0.953	0.995	1

Table S15. Block-size curve: five-fold CV macro-F1 by contiguous block size

Model	Block size	Purge gap	Macro-F1	Accuracy
1-NN (raw)	1	0	0.990 ± 0.003	0.990
1-NN (raw)	5	0	0.917 ± 0.010	0.916
1-NN (raw)	5	50	0.402 ± 0.057	0.505
1-NN (raw)	25	0	0.783 ± 0.020	0.783
1-NN (raw)	25	50	0.685 ± 0.040	0.691
1-NN (raw)	100	0	0.731 ± 0.032	0.735
1-NN (raw)	100	50	0.705 ± 0.027	0.712
1-NN (raw)	200	0	0.718 ± 0.024	0.720
1-NN (raw)	200	50	0.703 ± 0.022	0.706
1-NN (raw)	400	0	0.684 ± 0.052	0.690
1-NN (raw)	400	50	0.674 ± 0.056	0.681
LightGBM (raw+eng)	1	0	0.976 ± 0.003	0.976
LightGBM (raw+eng)	5	0	0.907 ± 0.006	0.907
LightGBM (raw+eng)	5	50	0.411 ± 0.057	0.514
LightGBM (raw+eng)	25	0	0.840 ± 0.015	0.842
LightGBM (raw+eng)	25	50	0.724 ± 0.023	0.734
LightGBM (raw+eng)	100	0	0.800 ± 0.034	0.807
LightGBM (raw+eng)	100	50	0.760 ± 0.048	0.773
LightGBM (raw+eng)	200	0	0.780 ± 0.054	0.786
LightGBM (raw+eng)	200	50	0.759 ± 0.045	0.766
LightGBM (raw+eng)	400	0	0.750 ± 0.065	0.757
LightGBM (raw+eng)	400	50	0.733 ± 0.079	0.742

Block size 5 with a 50-row purge leaves very little training data for classes 1–3; this setting is not interpretable.

Table S16. Windowed multichannel analysis of classes 1–3 (leave-one-block-out)

Model	Windows	Window (s)	Accuracy	Macro-F1	Kappa
LR	93	0.25	0.925	0.925	0.887
LightGBM	93	0.25	0.892	0.894	0.839

Features: per-channel log Welch band power (0.5–4, 4–8, 8–13, 13–30, 30–70 Hz) and log line length (96 features).

S4 Ablation and feature definitions

Table S17. Feature-group ablation with LightGBM

Task	Protocol	Configuration	Features	CV macro-F1	Test macro-F1
4class	random	all (raw+eng)	73	0.968 ± 0.005	0.975
4class	random	raw only	16	0.974 ± 0.004	0.982
4class	random	engineered only	57	0.901 ± 0.008	0.916
4class	random	only time	24	0.807 ± 0.019	0.817
4class	random	eng minus time	33	0.875 ± 0.014	0.878
4class	random	only freq	14	0.776 ± 0.018	0.773
4class	random	eng minus freq	43	0.861 ± 0.009	0.878
4class	random	only nonlinear	7	0.602 ± 0.009	0.596

4class	random	eng minus nonlinear	50	0.897 ± 0.005	0.907
4class	random	only wavelet	12	0.760 ± 0.007	0.777
4class	random	eng minus wavelet	45	0.881 ± 0.011	0.886
4class	blocked	all (raw+eng)	73	0.753 ± 0.066	0.798
4class	blocked	raw only	16	0.763 ± 0.054	0.807
4class	blocked	engineered only	57	0.634 ± 0.048	0.681
4class	blocked	only time	24	0.604 ± 0.032	0.636
4class	blocked	eng minus time	33	0.645 ± 0.043	0.682
4class	blocked	only freq	14	0.592 ± 0.028	0.613
4class	blocked	eng minus freq	43	0.622 ± 0.034	0.661
4class	blocked	only nonlinear	7	0.527 ± 0.025	0.516
4class	blocked	eng minus nonlinear	50	0.642 ± 0.046	0.699
4class	blocked	only wavelet	12	0.597 ± 0.029	0.646
4class	blocked	eng minus wavelet	45	0.603 ± 0.044	0.656
3class	random	all (raw+eng)	73	0.959 ± 0.006	0.963
3class	random	raw only	16	0.967 ± 0.007	0.968
3class	random	engineered only	57	0.864 ± 0.012	0.879
3class	random	only time	24	0.739 ± 0.014	0.755
3class	random	eng minus time	33	0.825 ± 0.009	0.827
3class	random	only freq	14	0.692 ± 0.013	0.690
3class	random	eng minus freq	43	0.816 ± 0.019	0.822
3class	random	only nonlinear	7	0.454 ± 0.014	0.480
3class	random	eng minus nonlinear	50	0.862 ± 0.012	0.860
3class	random	only wavelet	12	0.675 ± 0.014	0.697
3class	random	eng minus wavelet	45	0.839 ± 0.012	0.840
3class	blocked	all (raw+eng)	73	0.696 ± 0.035	0.724
3class	blocked	raw only	16	0.724 ± 0.030	0.750
3class	blocked	engineered only	57	0.561 ± 0.047	0.592
3class	blocked	only time	24	0.490 ± 0.033	0.505
3class	blocked	eng minus time	33	0.548 ± 0.050	0.574
3class	blocked	only freq	14	0.476 ± 0.047	0.482
3class	blocked	eng minus freq	43	0.525 ± 0.030	0.555
3class	blocked	only nonlinear	7	0.360 ± 0.019	0.362
3class	blocked	eng minus nonlinear	50	0.549 ± 0.044	0.592
3class	blocked	only wavelet	12	0.481 ± 0.033	0.527
3class	blocked	eng minus wavelet	45	0.530 ± 0.060	0.535

Table S18. Engineered feature names by group

Group	Count	Features
time	24	t_mean, t_std, t_min, t_max, t_range, t_median, t_iqr, t_skew, t_kurt, t_rms, t_meanabs, t_logenergy, t_hjorth_act, t_hjorth_mob, t_hjorth_comp, t_linelength, t_mad1, t_mad2, t_maxabsdiff, t_zc_mean, t_zc_raw, t_n_extrema, t_lag1_ac, t_norm_ll
freq	14	f_bin1, f_bin2, f_bin3, f_bin4, f_bin5, f_bin6, f_bin7, f_bin8, f_logpower, f_spec_entropy, f_centroid, f_edge80, f_peak_bin, f_low_high_ratio
nonlinear	7	n_perm_entropy, n_sample_entropy, n_app_entropy, n_svd_entropy, n_petrosian_fd, n_katz_fd, n_higuchi_fd
wavelet	12	w_A2_logenergy, w_A2_relenergy, w_A2_std, w_A2_meanabs, w_D2_logenergy, w_D2_relenergy, w_D2_std, w_D2_meanabs, w_D1_logenergy, w_D1_relenergy, w_D1_std, w_D1_meanabs

Table S19. Top-20 features by mean |SHAP|, three-class task, blocked protocol

Feature	GS	FS	SE	mean_abs_shap	gain_importance
X11	1.926	1.193	0.122	1.080	2962.821
X8	0.377	0.757	1.416	0.850	3290.217
X15	0.206	1.085	0.117	0.469	1331.376
X12	0.302	0.685	0.290	0.425	1319.484
X6	0.566	0.465	0.205	0.412	1507.080
X5	0.505	0.484	0.138	0.376	1754.900
t_hjorth_comp	0.151	0.145	0.783	0.360	1041.043
w_D1_std	0.405	0.580	0.044	0.343	2552.991
X9	0.151	0.408	0.462	0.340	1767.332
X3	0.626	0.272	0.106	0.335	860.559
X16	0.554	0.163	0.229	0.316	898.128
X1	0.237	0.143	0.522	0.301	1268.298

t_hjorth_mob	0.660	0.163	0.071	0.298	738.214
X4	0.324	0.445	0.120	0.296	1116.365
X14	0.383	0.327	0.148	0.286	745.136
X7	0.452	0.146	0.178	0.259	1014.838
X2	0.337	0.188	0.193	0.239	690.598
t_iqr	0.457	0.157	0.076	0.230	2559.147
X10	0.183	0.318	0.146	0.215	916.069
n_katz_fd	0.289	0.101	0.249	0.213	753.938

S5 Additional figures

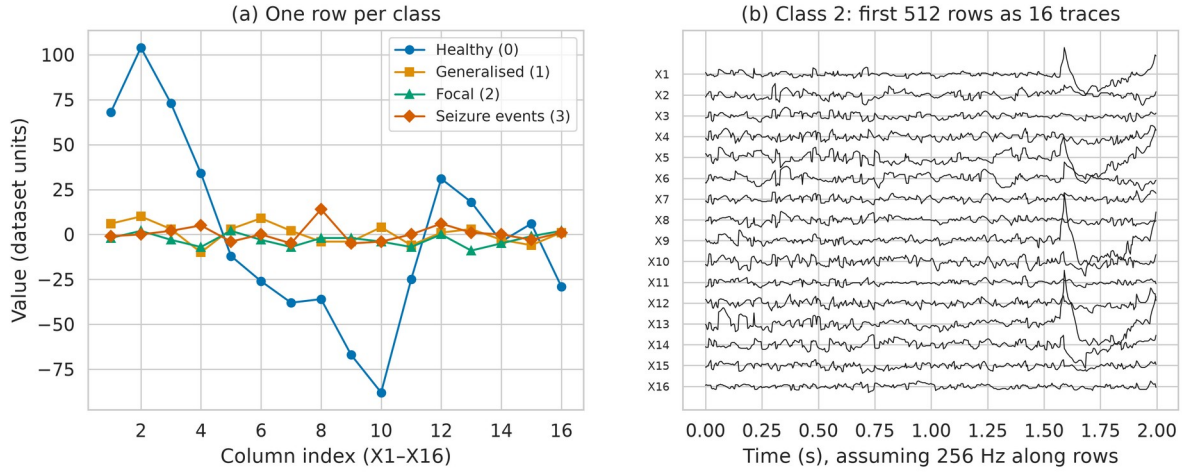


Figure S1. Example rows per class and class-2 rows displayed as 16 traces. (a) One row of each class plotted against column index. (b) The first 512 rows of class 2 plotted as 16 channels over time (2 s at 256 Hz), showing smooth evolution across rows, in contrast with class 0.

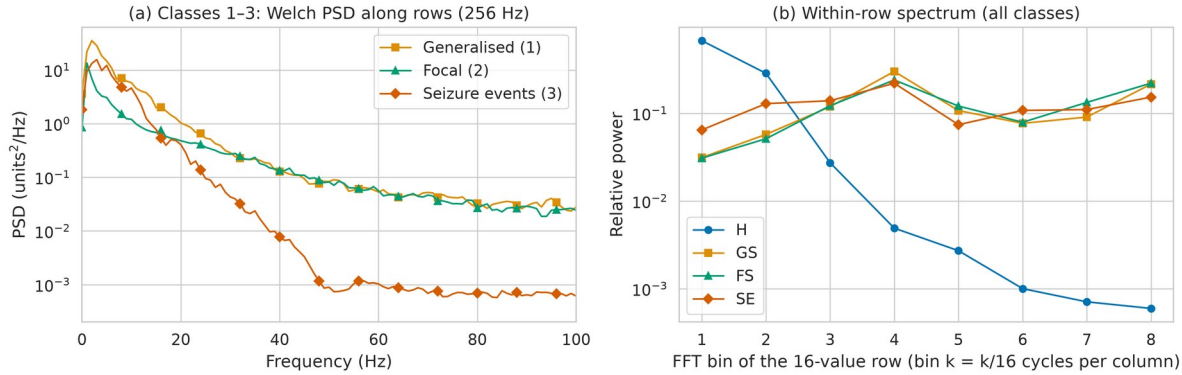


Figure S2. Spectral content of the BEED classes. (a) Welch power spectral density along rows for classes 1–3 (assuming 256 Hz), averaged over channels. (b) Relative power in the eight non-DC bins of the 16-value within-row FFT for all classes.

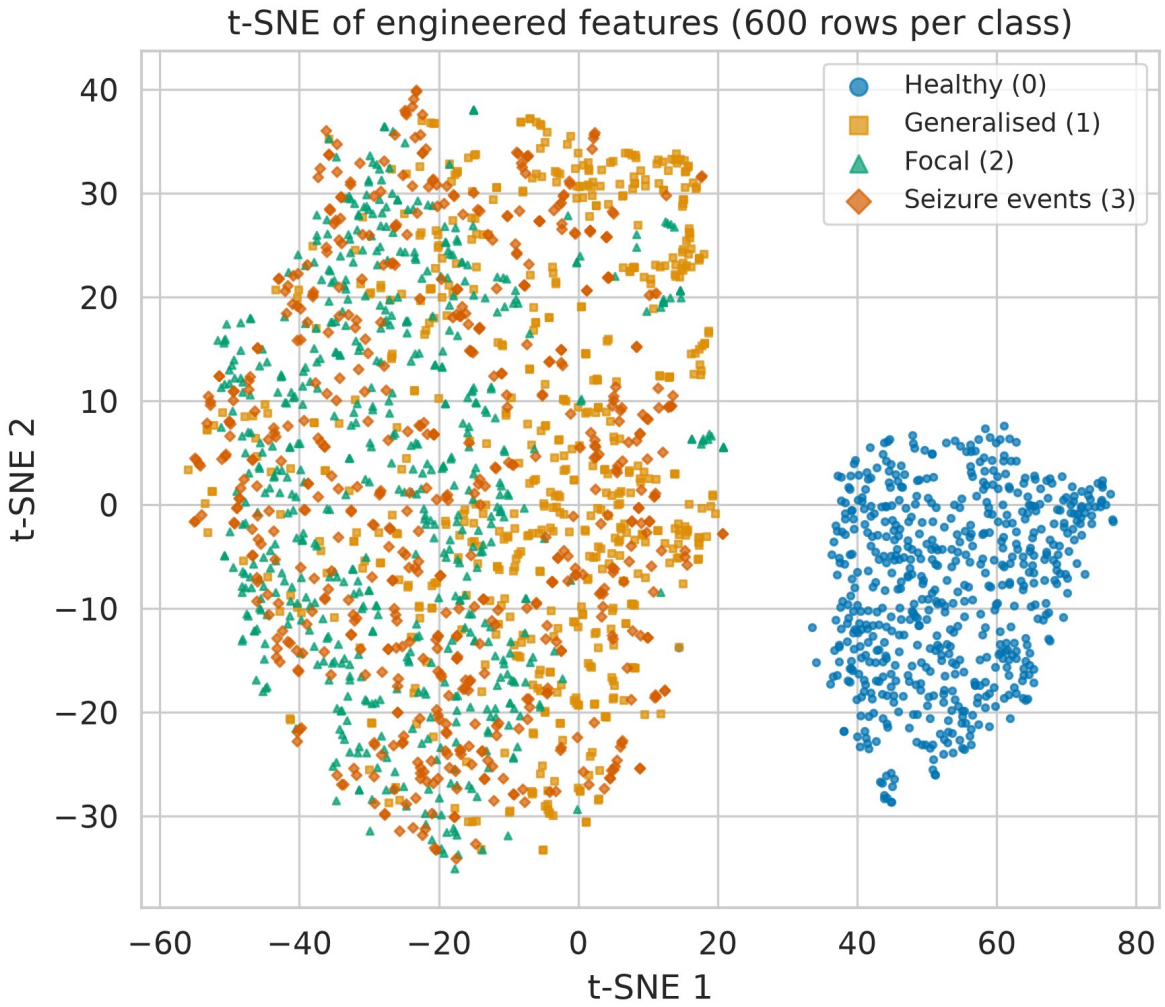


Figure S3. t-SNE embedding of engineered features. 600 randomly selected rows per class; class 0 forms a separate cluster, consistent with its different source.

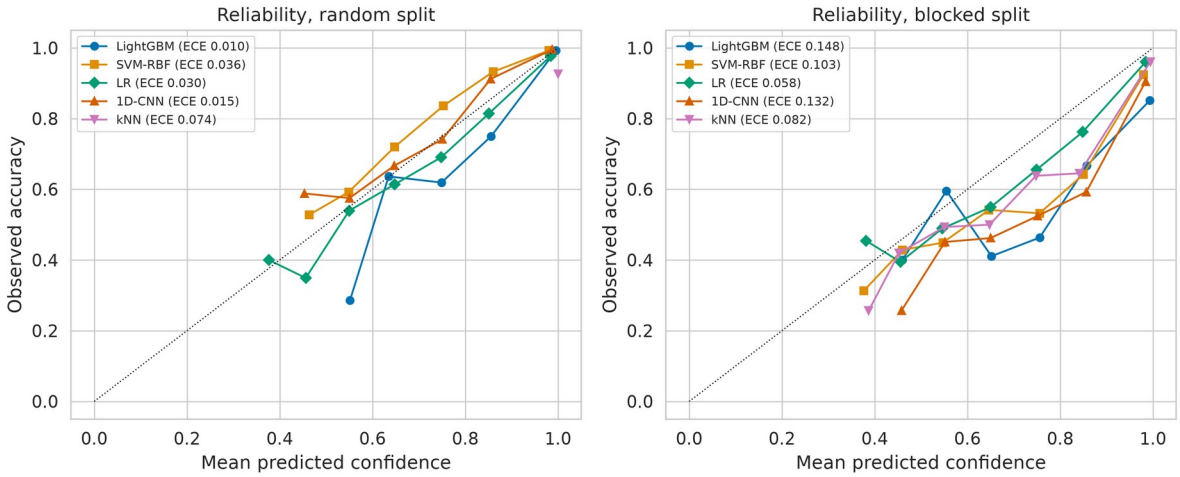


Figure S4. Reliability diagrams on the four-class hold-out test sets. Observed accuracy versus mean confidence in ten bins (bins with fewer than ten samples omitted); legend gives ECE.

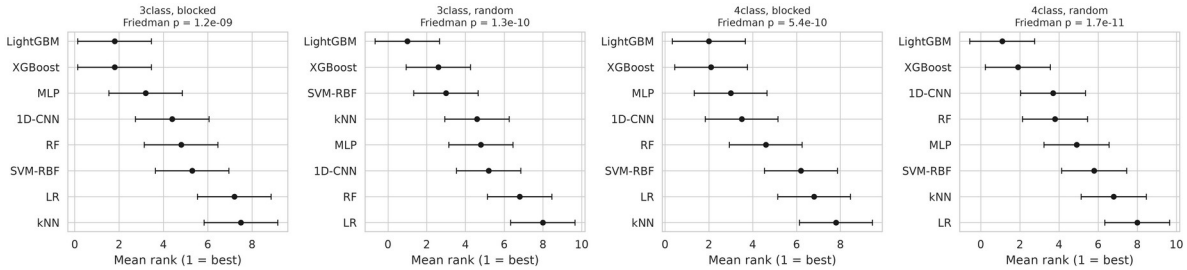


Figure S5. Mean ranks of models across repeated cross-validation folds. Error bars span the Nemenyi critical difference; models whose bars overlap are not significantly different.

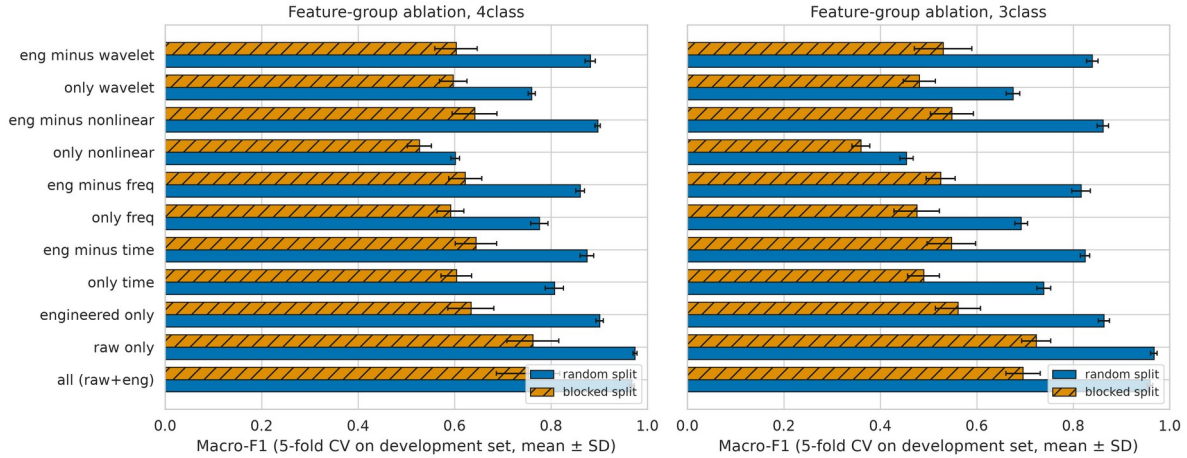


Figure S6. Feature-group ablation under random and blocked protocols. LightGBM with fixed hyper-parameters; five-fold CV on the development set (mean ± SD).

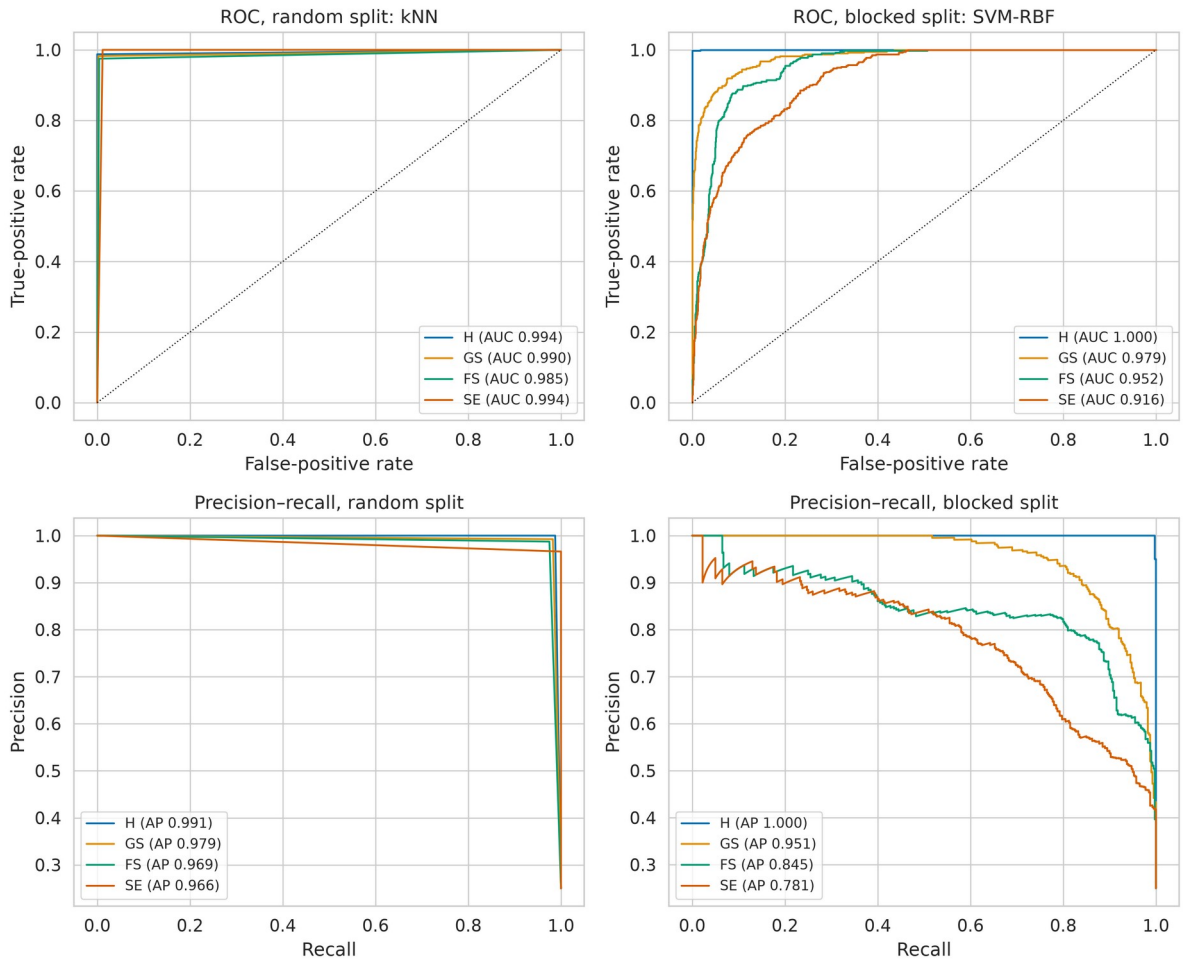


Figure S7. One-vs-rest ROC and precision–recall curves of the best four-class models. Left: random protocol; right: blocked protocol. AUC, area under the curve; AP, average precision.

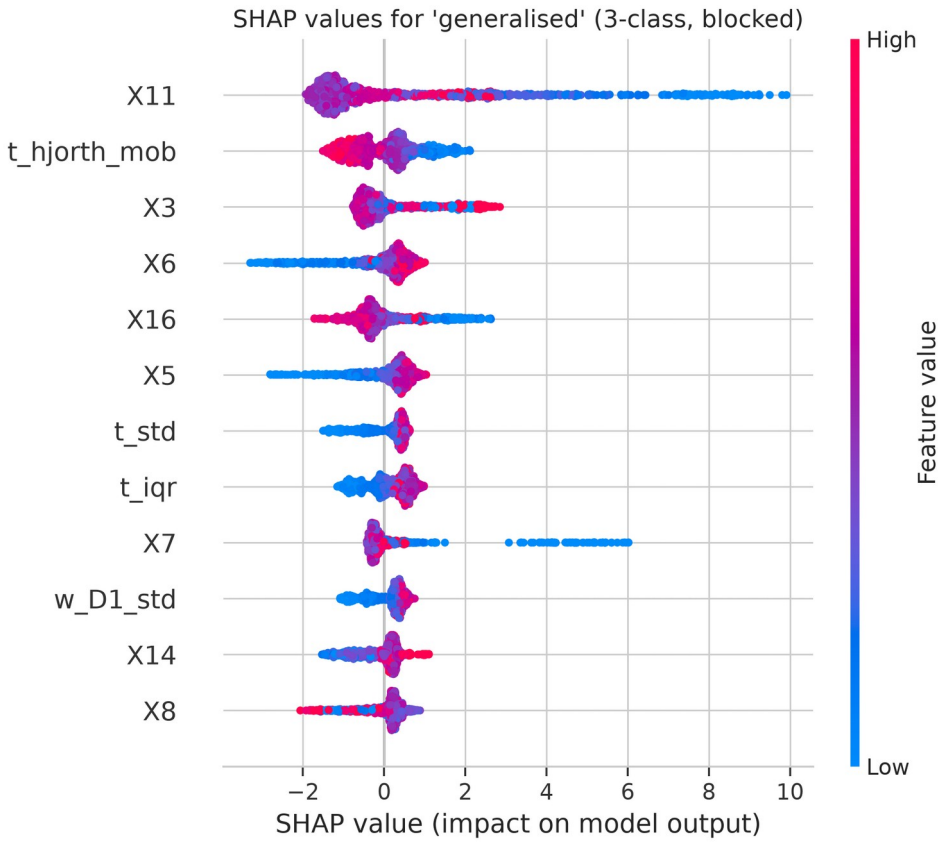


Figure S8. SHAP beeswarm for the generalised-seizure output (three-class, blocked). Each point is one test row; colour encodes feature value (red high, blue low).

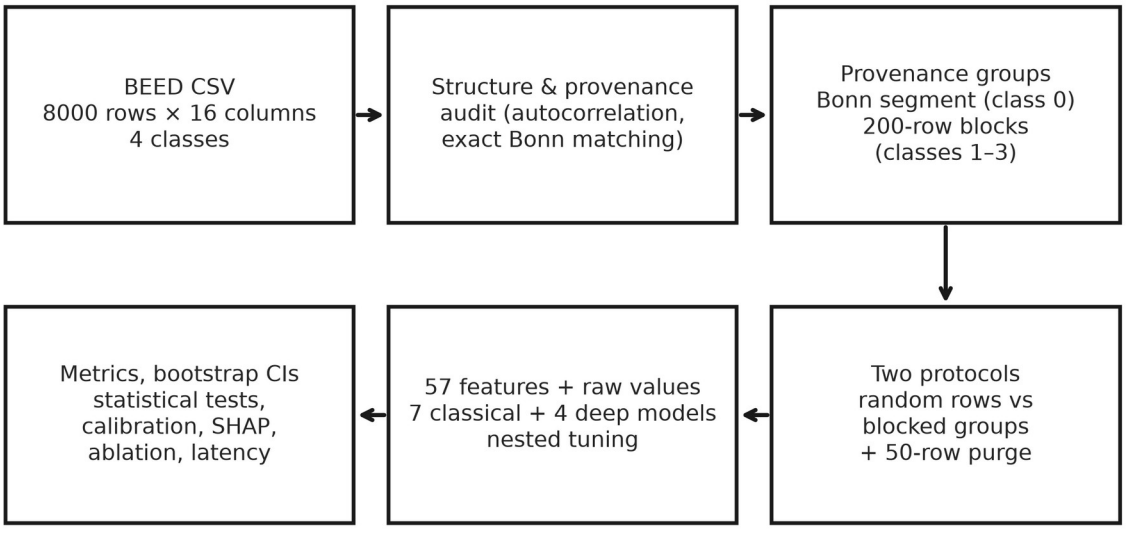


Figure S9. Overview of the audit and leakage-aware evaluation pipeline. Rows are grouped by provenance (Bonn segment for class 0; contiguous 200-row blocks for classes 1–3) before any split. Both protocols use the same features, models, nested tuning and metrics.

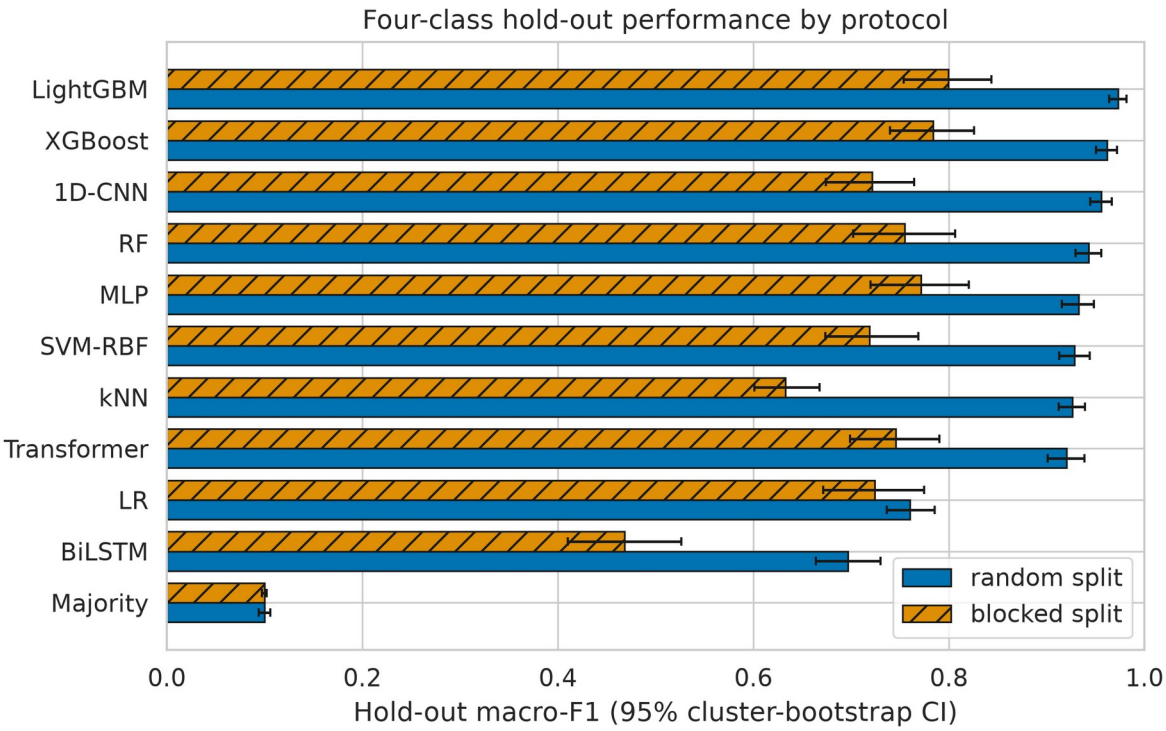


Figure S10. Four-class hold-out macro-F1 under random and blocked protocols. Classical models on raw+engineered features; deep models on raw values. Error bars: 95% cluster-bootstrap CIs. Hatched bars: blocked protocol. The same values are tabulated in Table 3 of the paper and Supplementary Table S2.